

# Project 2: Voronoï Diagrams, Optimal Transport and 2D Fluid Simulation

Labs 6, 7 and 8

Youssef Seghrouchni

June 2026

## 1 Introduction

This report covers Labs 6 to 8, which together implement a Lagrangian free-surface fluid simulator in C++. Lab 6 builds a Voronoï diagram by clipping convex polygons against half-planes. Lab 7 adds per-site weights to shift the bisectors (Laguerre cells) and uses L-BFGS to equalize cell areas via optimal transport. Lab 8 extends the OT problem with an extra air variable, clips each fluid cell to a disk to get a free surface, and runs a particle simulation under gravity.

## 2 Lab 6: Voronoï Diagram by Polygon Clipping

Given  $n$  sites  $P_1, \dots, P_n$  in the unit square, the Voronoï cell of  $P_i$  collects every point closer to  $P_i$  than to any other site:

$$V_i = \{x \in [0, 1]^2 : \|x - P_i\|^2 \leq \|x - P_j\|^2, \forall j \neq i\}.$$

Because each of those constraints is linear in  $x$ ,  $V_i$  is a convex polygon. The computation in `VoronoiDiagram::compute` starts each cell as the unit square and clips it one bisector at a time until all  $n - 1$  neighbours have been processed.

The clipping step is handled by `clip_by_bisector`. For a site pair  $(P_0, P_j)$  with optional weights  $(w_0, w_j)$ , a point  $X$  belongs to cell 0 when

$$\phi(X) = 2\langle X, P_j - P_0 \rangle - (\|P_j\|^2 - \|P_0\|^2) - w_0 + w_j \leq 0.$$

The routine then walks each edge  $A \rightarrow B$  of the current polygon. Both endpoints inside: keep  $B$ . Only  $A$  inside: output the crossing  $A + t(B - A)$ ,  $t = \phi(A)/(\phi(A) - \phi(B))$ . Only  $B$  inside: output the crossing then  $B$ . Neither inside: output nothing. Setting  $w_0 = w_j = 0$  recovers the standard Voronoï bisector; nonzero weights are the only addition needed for Lab 7. Each site is handled independently, so the outer loop is parallelised with `#pragma omp parallel for schedule(dynamic)`.

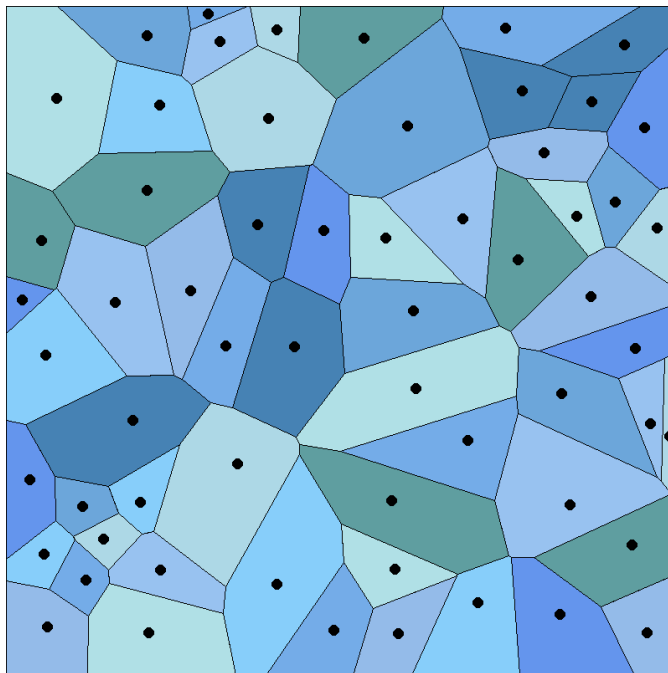


Figure 1: Voronoï diagram from Lab 6. Each coloured polygon is one cell, the black boundaries are the bisectors shared between neighbours, and the black dots are the sites  $P_i$ .

### 3 Lab 7: Semi-Discrete Optimal Transport

Giving each site  $P_i$  a scalar weight  $w_i$  shifts its bisectors, producing a *Laguerre cell*:

$$L_i(w) = \{x \in [0, 1]^2 : \|x - P_i\|^2 - w_i \leq \|x - P_j\|^2 - w_j, \forall j \neq i\}.$$

The code change is minimal: the constant inside  $\phi(X)$  gains two weight terms and the rest of the clipping is unchanged.

The goal is to find  $w$  such that  $|L_i(w)| = \lambda_i$  for all  $i$ , where  $\lambda_i = 1/n$  for a uniform target. This is a semi-discrete  $W_2$  optimal transport problem. The Kantorovich dual  $g(w)$  is concave, so minimising  $\mathcal{F} = -g$  is convex and L-BFGS converges to the global solution:

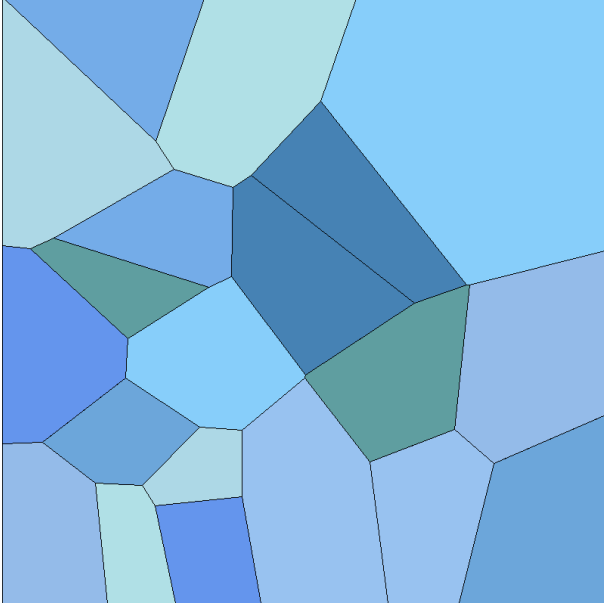
$$\mathcal{F}(w) = \sum_i \left[ w_i (|L_i| - \lambda_i) - \int_{L_i} \|x - P_i\|^2 dx \right], \quad \frac{\partial \mathcal{F}}{\partial w_i} = |L_i| - \lambda_i.$$

When every partial derivative is zero, each cell has exactly area  $\lambda_i$ . The polygon integrals (area via the shoelace formula, centroid, and the squared-distance integral) are all evaluated analytically from the vertex coordinates using standard cross-product identities.

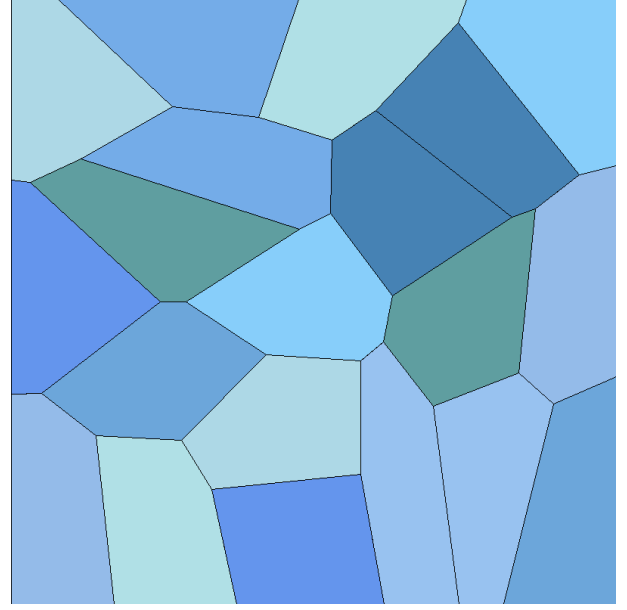
I tested this with  $n = 20$  random sites and target  $\lambda_i = 0.05$ . Running L-BFGS for up to 80 iterations (tolerance  $\varepsilon = 10^{-5}$ ) brought the maximum area error down to  $6.83 \times 10^{-6}$ , which confirms the solver is working.

| Sites $n$ | Target area | Max area error        |
|-----------|-------------|-----------------------|
| 20        | 0.05        | $6.83 \times 10^{-6}$ |

Table 1: Convergence of the L-BFGS optimal transport solver for  $n = 20$ .



(a) All weights zero: standard Voronoi diagram. Black dots are the sites  $P_i$ . Cell areas differ a lot.



(b) After L-BFGS: weights adjusted so every cell covers exactly  $1/20$  of the square (max error  $6.8 \times 10^{-6}$ , Table 1). Sites stay fixed.

Figure 2: Laguerre diagrams for the same 20 sites before and after optimal transport. Black dots mark the sites; colours label the same site in both views. Only the weights change, shifting the bisectors; the sites themselves do not move.

## 4 Lab 8: Fluid Simulation

The fluid is represented by  $N = 100$  Lagrangian particles. At every time step, the same OT solver from Lab 7 runs, but with one extra “air” site appended to the diagram ( $N + 1$  weights total). The prescribed masses are

$$\lambda_i = \frac{V_{\text{fluid}}}{N} \quad (\text{fluid}), \quad \lambda_{\text{air}} = 1 - V_{\text{fluid}} \quad (\text{air}),$$

where  $V_{\text{fluid}} = \pi R^2$  with  $R = 0.23$ . After solving, each fluid Laguerre cell  $L_i$  is clipped to a disk of radius  $r_i = \sqrt{w_i - w_{\text{air}}}$ . A particle with  $w_i \leq w_{\text{air}}$  gets an empty cell (it is considered air). This is what creates the free surface: the visible fluid region is the union of the disk-clipped cells, not the whole domain.

The 40-sided polygon approximation of each disk is clipped using `clip.by_edge`, which applies the same Sutherland-Hodgman walk as Lab 6, with the half-plane direction given by each polygon edge  $u \rightarrow v$ . One subtlety with the partial OT is the gradient for the air weight. Unlike the fluid gradients, it needs a sign flip because increasing  $w_{\text{air}}$  shrinks all disk radii at once and reduces total fluid area:

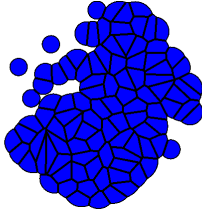
$$\frac{\partial \mathcal{F}}{\partial w_{\text{air}}} = \sum_i \lambda_i - \sum_i |C_i|,$$

where  $C_i$  denotes the disk-clipped cell. Getting this right was one of the harder parts of the lab.

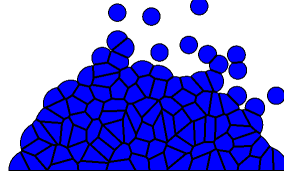
After the OT step, each particle is moved using a spring-gravity force:

$$F_i = m \mathbf{g} + \frac{1}{\varepsilon^2} (c_i - P_i),$$

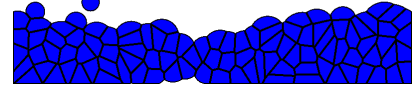
with  $\mathbf{g} = (0, -13.5)$ ,  $m = 200$ ,  $\varepsilon^2 = (0.004)^2$ , and  $c_i$  the centroid of the clipped cell. The spring pulls each particle toward the centre of its own fluid region, which keeps the blob from collapsing. Velocity and position update with forward Euler at  $\Delta t = 0.0024$ , and particles that exit  $[0, 1]^2$  bounce back with a 0.55 damping on the normal velocity. Rendering uses `save_frame_fluide`, which draws only the disk-clipped cells in blue; drawing full Laguerre cells produced distracting lines across the air region.



(a) Frame 0: particles released near  $(0.5, 0.75)$  and already falling. The circular shape of each disk-clipped cell is visible.



(b) Frame 60: blob has hit the floor and is spreading outward. Some particles detach and form droplets above.



(c) Frame 160: most of the fluid has settled as a thin layer along the bottom, with a few isolated droplets still airborne.

Figure 3: Three frames from the Lab 8 simulation ( $N = 100$ ,  $\Delta t = 0.0024$ , 180 frames total saved as `water_sim_k.png`). Blue: disk-clipped fluid cells. White: air. Black: boundaries between adjacent fluid cells.

## 5 Conclusion

What struck me about this project is how much mileage you get from a single geometric primitive. The polygon clipper written in Lab 6 is reused as-is in Labs 7 and 8, bisectors and disk edges alike. Moving from Voronoï to Laguerre cells is literally a one-line change in the half-plane formula; the free surface comes for free from the disk radius  $r_i = \sqrt{w_i - w_{\text{air}}}$ , with no separate surface mesh needed. The two things that needed the most attention were the sign of the air gradient in the partial OT and tuning the spring stiffness  $1/\varepsilon^2$  so the blob spreads realistically without the integrator blowing up.

## Course Feedback

The lab structure worked well for me. Each session introduced one new piece on top of what was already there, so the final simulator never felt overwhelming. I think the trickiest theoretical part was understanding why the Kantorovich dual is concave, and therefore why L-BFGS is applicable at all here.

## Declaration

This is my own work, even if i got a slightly little bit help from Neyl Nasr. The L-BFGS solver and `stb_image_write.h` were provided as course template files.